System Capacity & Care Load Analytics

HHS Unaccompanied Alien Children Program

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Executive Summary

Overall System Status: **HIGH**

Significant concern. Activate contingency plans and increase discharge processing capacity.

Dataset Overview

Dataset Overview

Metric	Value
Total Records	731
Date Range Start	2023-01-01 00:00:00
Date Range End	2024-12-31 00:00:00
Total Columns	52

Data Columns

Column	Data Type
Date	datetime64[ns]
CBP Custody	int64
Transfers	int64
HHS Care	int64
Discharges	int64
Total_System_Load	int64
Net_Daily_Intake	int64
Daily_Growth_Rate	float64
Backlog_Indicator	int64
Discharge_Offset_Ratio	float64
CBP Custody_Rolling_Mean_7d	float64
CBP Custody_Rolling_Std_7d	float64
CBP Custody_Rolling_Var_7d	float64
Transfers_Rolling_Mean_7d	float64
Transfers_Rolling_Std_7d	float64
Transfers_Rolling_Var_7d	float64
HHS Care_Rolling_Mean_7d	float64
HHS Care_Rolling_Std_7d	float64
HHS Care_Rolling_Var_7d	float64
Discharges_Rolling_Mean_7d	float64
Discharges_Rolling_Std_7d	float64
Discharges_Rolling_Var_7d	float64
Total_System_Load_Rolling_Mean_7d	float64
Total_System_Load_Rolling_Std_7d	float64
Total_System_Load_Rolling_Var_7d	float64

Column	Data Type
Net_Daily_Intake_Rolling_Mean_7d	float64
Net_Daily_Intake_Rolling_Std_7d	float64
Net_Daily_Intake_Rolling_Var_7d	float64
CBP Custody_Rolling_Mean_14d	float64
CBP Custody_Rolling_Std_14d	float64
CBP Custody_Rolling_Var_14d	float64
Transfers_Rolling_Mean_14d	float64
Transfers_Rolling_Std_14d	float64
Transfers_Rolling_Var_14d	float64
HHS Care_Rolling_Mean_14d	float64
HHS Care_Rolling_Std_14d	float64
HHS Care_Rolling_Var_14d	float64
Discharges_Rolling_Mean_14d	float64
Discharges_Rolling_Std_14d	float64
Discharges_Rolling_Var_14d	float64
Total_System_Load_Rolling_Mean_14d	float64
Total_System_Load_Rolling_Std_14d	float64
Total_System_Load_Rolling_Var_14d	float64
Net_Daily_Intake_Rolling_Mean_14d	float64
Net_Daily_Intake_Rolling_Std_14d	float64
Net_Daily_Intake_Rolling_Var_14d	float64
Week_Number	UInt32
Month	int32
Quarter	int32
Year	int32
Day_of_Week	int32
Weekend_Flag	bool

Data Validation Report

Data Validation Summary

Overall Validation Status: **PASS**

Key Performance Indicators

KPI	Value
Total Children Under Care	5097.0
Average Daily Intake	491.1491608071135
Average Daily Discharges	432.9630942457661
Average Bed Occupancy	3926.0820934337
Average Length of Stay	8.151.1493884404925
Net Intake Pressure	58.1.4559063994528
Care Load Volatility	0.4467841048394460083
Backlog Accumulation Rate	65.971.1283287671233
Discharge Efficiency	0.8941.278489431929
Average System Load	57462.0314637483

Healthcare Analytics Summary

Overall System Status: **HIGH**

Recommendation: Significant concern. Activate contingency plans and increase discharge processing capacity.

Warnings:

- Capacity stress detected

- Backlog increasing

- Rapid growth detected

Key Analytics Results

Metric	Value
Capacity Stress Status	STRESS
Stress Days	158
Backlog Trend	INCREASING
Current Backlog	47785
Net Intake Trend	INCREASING
Growth Status	RAPID_GROWTH
Total Growth %	532.9%

Recommendations

Significant concern. Activate contingency plans and increase discharge processing capacity.

Specific Actions:

1. Implement immediate capacity expansion measures and consider temporary overflow facilities.

2. Increase discharge processing capacity and review placement procedures to reduce backlog accumulation.

3. Review intake forecasting models and ensure adequate staffing levels for projected growth.

Conclusion

This report provides a comprehensive analysis of the HHS Unaccompanied Alien Children Program capacity and care load metrics. The current system status is **HIGH**. The analysis includes data validation, feature engineering, KPI calculations, and specialized healthcare analytics covering capacity stress, backlog trends, pressure windows, and growth patterns. Continuous monitoring and regular analysis of these metrics will help ensure optimal system performance and enable proactive response to capacity challenges.